Supplementary Material for "exdqlm: An R Package for Estimation and Analysis of Flexible Dynamic Quantile Linear Models"

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Abstract

This supplement derives the posterior targets, full conditional distributions, variational updates, Laplace–delta approximations, and numerical blocks used by the algorithms in **exdqlm**. It covers dynamic AL/DQLM and exAL/exDQLM state-space models, transfer-function state augmentation, static AL/exAL regression, ridge and Nishimura–Suchard regularized horseshoe shrinkage priors, forecasting, diagnostics, and posterior-predictive synthesis. The notation follows the main article, and each algorithmic block is linked to the corresponding exported package function.

Keywords: Bayesian quantile regression, dynamic linear models, asymmetric Laplace likelihood, extended asymmetric Laplace likelihood, MCMC, Laplace–delta variational Bayes.

S1. Scope and relation to the main article

The main article describes the user-facing workflow for **exdqlm**: dynamic exDQLM and DQLM fitting, transfer-function models, static exAL regression, forecasting, diagnostics, and posterior-predictive synthesis. This supplement gives the corresponding posterior targets, full conditional distributions, variational updates, Laplace–delta approximations, ELBO components, and numerical routines used by the package. Readers can use this document to verify the posterior targets, update equations, and diagnostic quantities behind the exported functions. The derivations build on the AL mixture representation of Kozumi and Kobayashi (2011), dynamic quantile linear models (Gonçalves, Migon, and Bastos 2017), the exAL/exDQLM formulation (Yan, Zheng, and Kottas 2025; Barata, Prado, and Sansó 2022), Gaussian state-space filtering, smoothing, and FFBS recursions described in Section S2.2, non-conjugate variational approximations (Wang and Blei 2013), and the `rhs_ns` prior (Nishimura and Suchard 2023).

Table S1 gives the main navigation map. The model sections derive the fitting targets and updates, Section S9 collects reusable numerical identities, and Table S4 links the exported functions to the corresponding computational blocks.

The notation follows the main article. In particular, $\mathbf{F}_t$ denotes the observation vector, $\mathbf{G}_t$

the state evolution matrix, $\mathbf{W}_t$ the evolution covariance, $\boldsymbol{\theta}_t$ the dynamic state, p_0 the target quantile level, and γ the exAL skewness parameter. Dimensions are stated when each object is introduced.

Main article topic	Main article location	Supplement location
exAL mixture and exDQLM hierarchy	Equations (1)–(6)	S2–S4
AL/DQLM special case	Section 2.5	S3
Discount-factor evolution	Section 2.3	S3.1, S4.1
Gaussian state updates	Section 2.2	S2.2
Transfer-function exDQLM	Equations (7)–(10)	S5
Static AL/exAL regression	Section 2.5, Example 4	S6–S7
Forecasting	Equation (16)	S8.1
Diagnostics	Equations (11)–(15)	S8.2
Posterior-predictive synthesis	Example 1	S8.3
Numerical blocks	Section 2.2	S9

Table S1: Crosswalk between the main article, this supplement, and the corresponding derivation sections.

S2. Notation and distributional conventions

Let $p_0 \in (0,1)$ be the target quantile. The response is scalar. In dynamic models, observations are indexed by $t = 1, \ldots, T$; in static regression, observations are indexed by $i = 1, \ldots, n$. For any positive definite matrix $\mathbf{M}$, $|\mathbf{M}|$ denotes its determinant.

S2.1. Core notation

The symbols p_γ, A_γ, B_γ, and C_γ are deterministic functions of (p_0, γ), whereas $\pi_\gamma(\gamma)$ denotes the prior density for γ on (L, U). Generic densities are denoted by $p(\cdot)$ only when no confusion with the exAL quantile map can arise. In variational sections, expectations without an explicit subscript are taken under the current mean-field distribution q. The symbol ξ is reserved for the `rhs_ns` half-Cauchy auxiliary variable; the transformed coordinate for γ in Laplace–delta calculations is denoted by z_γ.

The latent variables v_t, s_t, v_i, and s_i are introduced for posterior computation and are updated internally by the fitting routines; they are not user-supplied inputs. Practitioners specify the response, target quantile, model components, priors, and inference controls described in the main article.

The distributional parameterizations are also fixed throughout: $\mathcal{N}(m, V)$ uses variance V;

$$x \sim \mathrm{IG}(a, b) \quad \Longleftrightarrow \quad p(x) = \frac{b^a}{\Gamma(a)} x^{-a-1} \exp(-b/x), \qquad x > 0, \tag{S1}$$

equivalently $x^{-1} \sim \mathrm{Gamma}(a, b)$ under a shape-rate gamma parameterization. The exponential distribution $\mathrm{Exp}(\mathrm{rate} = 1/\sigma)$ has density $\sigma^{-1} \exp(-x/\sigma)$, matching the main article's mean-σ convention for the AL and exAL mixture variables. Finally, $\mathcal{N}^+(m, V)$ denotes

Symbol	Meaning
p_0	Target quantile level.
$p_\gamma = p(p_0, \gamma)$	Internal exAL quantile map defined in the main article.
A_{p_0}, B_{p_0}	AL constants $A(p_0)$ and $B(p_0)$.
$A_\gamma, B_\gamma, C_\gamma$	exAL constants evaluated at p_γ.
L, U	Lower and upper admissible bounds for γ.
$\pi_\gamma(\gamma)$	Prior density for γ on (L, U).
$\boldsymbol{\theta}_t, \mathbf{F}_t, \mathbf{G}_t, \mathbf{W}_t$	Dynamic state, observation vector, evolution matrix, and evolution covariance.
d_t, Q_t	Observation offset and variance in the generic Gaussian state update.
v_t, s_t	exAL latent variables in the dynamic model.
$\boldsymbol{\beta}, \mathbf{X}$	Static regression coefficients and design matrix.
v_i, s_i	exAL latent variables in the static model.
$\mathcal{A}$	Active coefficient set receiving `rhs_ns` shrinkage.
$\mathcal{L}(q)$	Evidence lower bound.

Table S2: Core notation used throughout the supplement. The shorthand p_γ, A_γ, B_γ, and C_γ refers to functions of (p_0, γ) defined in the main article.

$\mathcal{N}(m, V)$ truncated to $(0, \infty)$. The generalized inverse Gaussian parameterization is given in (S10). Throughout, $\phi(\cdot)$ and $\Phi(\cdot)$ denote the standard normal density and distribution function.

The asymmetric Laplace (AL) mixture representation used by the package is

$$v \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \qquad v > 0, \tag{S2}$$

$$y \mid \eta, \sigma, v \sim \mathcal{N}\{\eta + A(p_0)v,\ \sigma B(p_0)v\}, \tag{S3}$$

where

$$A(p) = \frac{1-2p}{p(1-p)}, \qquad B(p) = \frac{2}{p(1-p)}. \tag{S4}$$

The AL model is the $\gamma = 0$ special case of the exAL formulation below.

The exAL distribution is represented through an additional latent variable $s \sim \mathcal{N}^+(0,1)$, a standard normal truncated to $(0, \infty)$. For $\gamma \in (L, U)$, define

$$g(\gamma) = 2\Phi(-|\gamma|)\exp(\gamma^2/2), \tag{S5}$$

$$p_\gamma = \mathbb{I}(\gamma < 0) + \frac{p_0 - \mathbb{I}(\gamma < 0)}{g(\gamma)}, \tag{S6}$$

$$A_\gamma = A(p_\gamma), \qquad B_\gamma = B(p_\gamma), \qquad C_\gamma = \{\mathbb{I}(\gamma > 0) - p_\gamma\}^{-1}. \tag{S7}$$

The endpoints $L < 0 < U$ are the roots $g(L) = 1 - p_0$ and $g(U) = p_0$. Conditional on (v, s, σ, γ),

$$y \mid \eta, \sigma, \gamma, v, s \sim \mathcal{N}\left(\eta + C_\gamma \sigma |\gamma| s + A_\gamma v,\ \sigma B_\gamma v\right). \tag{S8}$$

The package uses an inverse-gamma prior $\sigma \sim \text{IG}(a_\sigma, b_\sigma)$ and a proper truncated Student-t prior $\pi_\gamma(\gamma)$ for γ on (L, U). If m_γ, s_γ, and ν_γ denote its location, scale, and degrees of freedom, then

$$\pi_\gamma(\gamma) = \frac{s_\gamma^{-1} t_{\nu_\gamma}\{(\gamma - m_\gamma)/s_\gamma\}}{T_{\nu_\gamma}\{(U - m_\gamma)/s_\gamma\} - T_{\nu_\gamma}\{(L - m_\gamma)/s_\gamma\}} \mathbb{I}(L < \gamma < U), \tag{S9}$$

where $t_\nu(\cdot)$ and $T_\nu(\cdot)$ are the standard Student-t density and distribution function.

The generalized inverse Gaussian distribution is parameterized as

$$x \sim \mathrm{GIG}(\lambda, \chi, \psi_{\mathrm{GIG}}) \quad \iff \quad p(x) \propto x^{\lambda-1} \exp\left\{-\frac{1}{2}\left(\frac{\chi}{x} + \psi_{\mathrm{GIG}} x\right)\right\}, \qquad x > 0. \tag{S10}$$

For $r \in \mathbb{R}$,

$$\mathbb{E}(x^r) = \left(\frac{\chi}{\psi_{\mathrm{GIG}}}\right)^{r/2} \frac{K_{\lambda+r}(\sqrt{\chi\psi_{\mathrm{GIG}}})}{K_\lambda(\sqrt{\chi\psi_{\mathrm{GIG}}})}, \tag{S11}$$

where $K_\nu(\cdot)$ is the modified Bessel function of the second kind.

S2.2. Gaussian state updates for dynamic models

All dynamic MCMC and VB updates for the state path reduce to the same conditionally Gaussian state-space calculation. The package uses the observation-offset representation

$$y_t = \mathbf{F}_t^\top \boldsymbol{\theta}_t + d_t + e_t, \qquad e_t \sim \mathcal{N}(0, Q_t), \tag{S12}$$

where d_t is the scalar observation offset and $Q_t > 0$ is the scalar conditional observation variance. In the implementation these quantities are stored as the extra observation offset and variance, respectively. Equivalently, with $z_t = y_t - d_t$,

$$z_t = \mathbf{F}_t^\top \boldsymbol{\theta}_t + e_t, \qquad e_t \sim \mathcal{N}(0, Q_t).$$

Together with

$$\boldsymbol{\theta}_0 \sim \mathcal{N}(\boldsymbol{m}_0, \mathbf{C}_0), \qquad \boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} \sim \mathcal{N}(\mathbf{G}_t \boldsymbol{\theta}_{t-1}, \mathbf{W}_t),$$

this defines the Gaussian state update used below.

Initialize $\boldsymbol{m}_{0|0} = \boldsymbol{m}_0$ and $\mathbf{C}_{0|0} = \mathbf{C}_0$. For $t = 1, \ldots, T$, the Kalman prediction and filtering recursions are

$$\boldsymbol{a}_t = \mathbf{G}_t \boldsymbol{m}_{t-1|t-1}, \qquad \mathbf{R}_t = \mathbf{G}_t \mathbf{C}_{t-1|t-1} \mathbf{G}_t^\top + \mathbf{W}_t, \tag{S13}$$

$$\mu_t^y = d_t + \mathbf{F}_t^\top \boldsymbol{a}_t, \qquad S_t = \mathbf{F}_t^\top \mathbf{R}_t \mathbf{F}_t + Q_t, \tag{S14}$$

$$\delta_t^y = y_t - \mu_t^y, \qquad \boldsymbol{k}_t = \mathbf{R}_t \mathbf{F}_t S_t^{-1}, \tag{S15}$$

$$\boldsymbol{m}_{t|t} = \boldsymbol{a}_t + \boldsymbol{k}_t \delta_t^y, \qquad \mathbf{C}_{t|t} = \mathbf{R}_t - \boldsymbol{k}_t S_t \boldsymbol{k}_t^\top. \tag{S16}$$

The same forward pass gives the Gaussian auxiliary normalizing constant

$$\log p_{\mathrm{G}}(\boldsymbol{y}) = \sum_{t=1}^T \log \mathcal{N}\{y_t \mid d_t + \mathbf{F}_t^\top \boldsymbol{a}_t, S_t\}, \tag{S17}$$

which is used in the ELBO state-block identities below.

For MCMC, the state path is sampled by forward-filtering backward-sampling (FFBS) (Carter and Kohn 1994; Frühwirth-Schnatter 1994). After the forward filter, draw

$$\boldsymbol{\theta}_T \sim \mathcal{N}(\boldsymbol{m}_{T|T}, \mathbf{C}_{T|T}).$$

For $t = T-1, \ldots, 0$, define

$$\mathbf{J}_t = \mathbf{C}_{t|t} \mathbf{G}_{t+1}^\top \mathbf{R}_{t+1}^{-1}. \tag{S18}$$

Then sample

$$\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t+1}, \boldsymbol{y} \sim \mathcal{N}(\boldsymbol{h}_t, \mathbf{H}_t), \tag{S19}$$

where

$$\boldsymbol{h}_t = \boldsymbol{m}_{t|t} + \mathbf{J}_t(\boldsymbol{\theta}_{t+1} - \boldsymbol{a}_{t+1}), \tag{S20}$$

$$\mathbf{H}_t = \mathbf{C}_{t|t} - \mathbf{J}_t \mathbf{R}_{t+1} \mathbf{J}_t^\top. \tag{S21}$$

For VB and LDVB, the state factor $q(\boldsymbol{\theta}_{0:T})$ is updated deterministically by the Rauch–Tung–Striebel backward smoother applied to the same Gaussian system. Set $\boldsymbol{m}_{T|T}$ and $\mathbf{C}_{T|T}$ equal to the final filtered moments. For $t = T-1, \dots, 0$,

$$\boldsymbol{m}_{t|T} = \boldsymbol{m}_{t|t} + \mathbf{J}_t(\boldsymbol{m}_{t+1|T} - \boldsymbol{a}_{t+1}), \tag{S22}$$

$$\mathbf{C}_{t|T} = \mathbf{C}_{t|t} + \mathbf{J}_t(\mathbf{C}_{t+1|T} - \mathbf{R}_{t+1})\mathbf{J}_t^\top. \tag{S23}$$

Thus MCMC uses stochastic backward sampling from the Gaussian conditional state posterior, whereas VB/LDVB uses deterministic smoothing to update the Gaussian factor $q(\boldsymbol{\theta}_{0:T})$. Posterior samples returned by the package from VB/LDVB fits are generated after convergence from the fitted variational factors; this sampling step is separate from the coordinate update.

The discount-factor construction changes how $\mathbf{W}_t$, or equivalently $\mathbf{R}_t$, is formed during the prediction step (West, Harrison, and Migon 1985). For example, scalar discounting can be written as

$$\mathbf{R}_t = \delta^{-1}\mathbf{G}_t \mathbf{C}_{t-1|t-1} \mathbf{G}_t^\top,$$

which is equivalent at that step to

$$\mathbf{W}_t = \frac{1-\delta}{\delta}\mathbf{G}_t \mathbf{C}_{t-1|t-1} \mathbf{G}_t^\top.$$

Componentwise discounting applies the same construction blockwise. Once the prediction covariance $\mathbf{R}_t$ is formed, the filtering, backward-sampling, and smoothing equations above are unchanged.

Table S3 gives the offset and variance used by each dynamic algorithm. The variational rows are obtained by completing the square in the expected Gaussian log likelihood.

S3. Dynamic DQLM under the AL likelihood

S3.1. Model hierarchy and posterior target

Let $\boldsymbol{\theta}_t \in \mathbb{R}^q$ be the latent state, $\mathbf{F}_t \in \mathbb{R}^q$ the observation vector, $\mathbf{G}_t \in \mathbb{R}^{q\times q}$ the evolution matrix, and $\mathbf{W}_t \in \mathbb{R}^{q\times q}$ a positive semidefinite evolution covariance. The dynamic DQLM is

$$\boldsymbol{\theta}_0 \sim \mathcal{N}(\boldsymbol{m}_0, \mathbf{C}_0), \qquad \boldsymbol{m}_0 \in \mathbb{R}^q,\ \mathbf{C}_0 \in \mathbb{R}^{q\times q}, \tag{S24}$$

$$\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} \sim \mathcal{N}(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}_t), \tag{S25}$$

$$v_t \mid \sigma \sim \mathrm{Exp}(\mathrm{rate} = 1/\sigma), \tag{S26}$$

$$y_t \mid \boldsymbol{\theta}_t, \sigma, v_t \sim \mathcal{N}(\mathbf{F}_t^\top\boldsymbol{\theta}_t + A_{p_0}v_t,\ \sigma B_{p_0} v_t), \tag{S27}$$

Dynamic update	Offset d_t	Variance Q_t	State update
DQLM MCMC	$A_{p_0} v_t$	$\sigma B_{p_0} v_t$	Kalman filter and FFBS backward sampling.
exDQLM MCMC	$A_\gamma v_t + C_\gamma \sigma \lvert\gamma\rvert s_t$	$\sigma B_\gamma v_t$	Kalman filter and FFBS backward sampling.
DQLM VB	$A_{p_0}/m_{1/v,t}$	$B_{p_0}/\{\kappa m_{1/v,t}\}$	Kalman filter and deterministic RTS smoothing.
exDQLM LDVB	$\{\kappa_2 + \kappa_4 \bar{s}_t m_{1/v,t}\}/\phi_t$	ϕ_t^{-1}, $\phi_t = \kappa_1 m_{1/v,t}$	Kalman filter and deterministic RTS smoothing.
Transfer-function exDQLM	Same as the selected base dynamic update	Same as the selected base dynamic update	Apply the same update after replacing $(\mathbf{F}_t, \mathbf{G}_t, \mathbf{W}_t, \boldsymbol{\theta}_t)$ by the augmented quantities in Section S5.

Table S3: Offset and variance representation used by the generic Gaussian state update in (S12). Here $\kappa = \mathbb{E}(\sigma^{-1})$ in the DQLM VB row, $m_{1/v,t} = \mathbb{E}(v_t^{-1})$, $\bar{s}_t = \mathbb{E}(s_t)$, and the κ_j moments are defined in (S60).

where $A_{p_0} = A(p_0)$ and $B_{p_0} = B(p_0)$. With $\boldsymbol{v} = (v_1, \ldots, v_T)^\top$ and $\boldsymbol{\theta}_{0:T} = (\boldsymbol{\theta}_0, \ldots, \boldsymbol{\theta}_T)$, the posterior target is

$$p(\boldsymbol{\theta}_{0:T}, \sigma, \boldsymbol{v} \mid \boldsymbol{y}) \propto p(\sigma)\, p(\boldsymbol{\theta}_0) \prod_{t=1}^{T} p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1}) \prod_{t=1}^{T} p(v_t \mid \sigma) \prod_{t=1}^{T} p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t). \tag{S28}$$

Equivalently, conditional on $(\sigma, \boldsymbol{v})$, the pseudo-observation $\tilde{y}_t = y_t - A_{p_0} v_t$ satisfies

$$\tilde{y}_t = \mathbf{F}_t^\top \boldsymbol{\theta}_t + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \sigma B_{p_0} v_t). \tag{S29}$$

Thus the conditional state posterior is the Gaussian state-space posterior defined in Section S2.2, with the DQLM MCMC offset and variance given in Table S3. The package constructs the evolution covariance through the discount-factor state-space rules described in the main article; once the prediction covariance is formed, the filtering and backward updates are those in Section S2.2.

S3.2. MCMC updates

One Gibbs sweep for the dynamic DQLM uses the following full conditionals.

State path. Given $(\sigma, \boldsymbol{v})$, sample $\boldsymbol{\theta}_{0:T}$ by the FFBS recursion in Section S2.2, using $d_t = A_{p_0} v_t$ and $Q_t = \sigma B_{p_0} v_t$.

Scale. Let $r_t = y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t$. Then

$$\sigma \mid \boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \boldsymbol{y} \sim \mathrm{IG}\left(a_\sigma + \frac{3T}{2},\ b_\sigma + \sum_{t=1}^{T} v_t + \sum_{t=1}^{T} \frac{(r_t - A_{p_0} v_t)^2}{2 B_{p_0} v_t} \right). \tag{S30}$$

Latent AL variables. For $t = 1, \ldots, T$,

$$v_t \mid \boldsymbol{\theta}_t, \sigma, y_t \sim \mathrm{GIG}\left(\frac{1}{2},\ \frac{r_t^2}{\sigma B_{p_0}},\ \frac{A_{p_0}^2}{\sigma B_{p_0}} + \frac{2}{\sigma} \right). \tag{S31}$$

Algorithm S1. Dynamic DQLM MCMC.

The MCMC kernel leaves the augmented posterior in (S28) invariant; posterior summaries are Monte Carlo estimates from the retained finite chain.

Input: observations $\boldsymbol{y}$, model matrices $(\mathbf{F}_t, \mathbf{G}_t, \mathbf{W}_t)$, initial state prior, AL prior hyperparameters, and MCMC controls N_{burn} and N_{mcmc}.

Output: retained posterior draws for $(\boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \sigma)$ and filtered/smoothed state summaries.

(1) Initialize $\boldsymbol{\theta}_{0:T}$, $\boldsymbol{v}$, and σ.

(2) For $m = 1, \ldots, N_{\text{burn}} + N_{\text{mcmc}}$:

   (a) sample $\boldsymbol{\theta}_{0:T}$ by FFBS using Section S2.2 and the DQLM MCMC row of Table S3;

   (b) sample σ from (S30);

   (c) sample each v_t from (S31);

   (d) if $m > N_{\text{burn}}$, retain the current draw.

(3) Return the retained draws and state summaries.

S3.3. Mean-field VB and ELBO

The dynamic DQLM VB approximation uses

$$q(\boldsymbol{\theta}_{0:T}, \sigma, \boldsymbol{v}) = q(\boldsymbol{\theta}_{0:T}) q(\sigma) \prod_{t=1}^{T} q(v_t), \tag{S32}$$

with $q(\sigma) = \text{IG}(a_q, b_q)$ and $q(v_t) = \text{GIG}(1/2, \chi_{v,t}, \psi_{\text{GIG},t})$. Define

$$\kappa = \mathbb{E}_q(\sigma^{-1}) = \frac{a_q}{b_q}, \qquad m_{1/v,t} = \mathbb{E}_q(v_t^{-1}), \qquad \bar{v}_t = \mathbb{E}_q(v_t). \tag{S33}$$

The state factor $q(\boldsymbol{\theta}_{0:T})$ is a joint Gaussian smoothing distribution, not a product over time. The latent-variable updates require only its marginal means and variances. The state factor is the Gaussian posterior of the auxiliary model

$$y_t^\star = \mathbf{F}_t^\top \boldsymbol{\theta}_t + \epsilon_t^\star, \qquad \epsilon_t^\star \sim \mathcal{N}(0, R_t^\star), \qquad y_t^\star = y_t - \frac{A_{p_0}}{m_{1/v,t}}, \quad R_t^\star = \frac{B_{p_0}}{\kappa m_{1/v,t}}. \tag{S34}$$

Equivalently, this is the generic observation-offset form in (S12) with $d_t = A_{p_0}/m_{1/v,t}$ and $Q_t = B_{p_0}/\{\kappa m_{1/v,t}\}$. After applying the deterministic smoother in Section S2.2, write $q(\boldsymbol{\theta}_t) = \mathcal{N}(\boldsymbol{m}_{t|T}, \mathbf{C}_{t|T})$ and define

$$\bar{r}_t = y_t - \mathbf{F}_t^\top \boldsymbol{m}_{t|T}, \tag{S35}$$

$$\overline{r_t^2} = \bar{r}_t^2 + \mathbf{F}_t^\top \mathbf{C}_{t|T} \mathbf{F}_t. \tag{S36}$$

The remaining coordinate-ascent variational inference (CAVI) updates are

$$q(v_t) = \mathrm{GIG}\left(\frac{1}{2},\ \frac{\kappa}{B_{p_0}}\overline{r_t^2},\ \kappa\left(2 + \frac{A_{p_0}^2}{B_{p_0}}\right)\right), \tag{S37}$$

$$q(\sigma) = \mathrm{IG}(a_q, b_q), \qquad a_q = a_\sigma + \frac{3T}{2}, \tag{S38}$$

$$b_q = b_\sigma + \sum_{t=1}^{T} \bar{v}_t + \frac{1}{2B_{p_0}} \sum_{t=1}^{T} \left\{ m_{1/v,t}\overline{r_t^2} - 2A_{p_0}\bar{r}_t + A_{p_0}^2 \bar{v}_t \right\}. \tag{S39}$$

Algorithm S2. Dynamic DQLM VB.
Input: observations $\boldsymbol{y}$, model matrices $(\mathbf{F}_t, \mathbf{G}_t, \mathbf{W}_t)$, initial state prior, AL prior hyperparameters, convergence tolerance, and optional posterior-sampling control `n.samp`.
Output: fitted variational factors for $(\boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \sigma)$, convergence diagnostics, ELBO values when requested, and optional approximate posterior draws.

(1) Initialize the variational parameters for $q(\boldsymbol{\theta}_{0:T})$, $q(\sigma)$, and $q(v_t)$.

(2) Repeat until the convergence criterion is met:

  (a) update $q(\boldsymbol{\theta}_{0:T})$ by the deterministic smoother in Section S2.2, using the DQLM VB row of Table S3;
  (b) update $q(v_t)$ for $t = 1, \dots, T$ using the GIG CAVI update above;
  (c) update $q(\sigma)$ using (S39);
  (d) monitor moment changes and, when ELBO monitoring is enabled, $\mathcal{L}(q)$.

(3) Draw approximate posterior samples from the fitted variational factors when requested by `n.samp`.

(4) Return the fitted factors, diagnostics, and optional draws.

In the package, this AL/DQLM path is selected in the dynamic fitting routines with `dqlm.ind = TRUE`.
The ELBO is

$$\mathcal{L}(q) = \mathbb{E}_q\{\log p(\boldsymbol{y}, \boldsymbol{\theta}_{0:T}, \sigma, \boldsymbol{v})\} - \mathbb{E}_q\{\log q(\boldsymbol{\theta}_{0:T}, \sigma, \boldsymbol{v})\} \tag{S40}$$

$$= \mathcal{L}_\theta + \mathbb{E}_q\{\log p(\sigma)\} + \sum_{t=1}^{T} \mathbb{E}_q\{\log p(v_t \mid \sigma)\} + \sum_{t=1}^{T} \mathbb{E}_q\{\log p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t)\} \tag{S41}$$

$$+ H\{q(\sigma)\} + \sum_{t=1}^{T} H\{q(v_t)\}. \tag{S42}$$

The state-path term $\mathcal{L}_\theta$ is evaluated from the auxiliary Gaussian state-space model in (S34) through the Kalman marginal likelihood identity

$$\mathcal{L}_\theta = \log p(\boldsymbol{y}^\star) - \mathbb{E}_q\{\log p(\boldsymbol{y}^\star \mid \boldsymbol{\theta}_{0:T})\}, \tag{S43}$$

where $p(\boldsymbol{y}^\star)$ is the marginal likelihood of the auxiliary Gaussian state-space model. This identity is equivalent to $\mathbb{E}_q\{\log p(\boldsymbol{\theta}_{0:T})\} + H\{q(\boldsymbol{\theta}_{0:T})\}$ for the current state update.

The remaining terms in (S42) are computable from the current variational moments. Let $\bar{\ell}_\sigma = \mathbb{E}(\log \sigma)$ and $\bar{\ell}_{v,t} = \mathbb{E}(\log v_t)$. Then

$$\mathbb{E}_q\{\log p(\sigma)\} = a_\sigma \log b_\sigma - \log \Gamma(a_\sigma) - (a_\sigma + 1)\bar{\ell}_\sigma - b_\sigma \kappa, \tag{S44}$$

$$\sum_{t=1}^{T} \mathbb{E}_q\{\log p(v_t \mid \sigma)\} = -T\bar{\ell}_\sigma - \kappa \sum_{t=1}^{T} \bar{v}_t, \tag{S45}$$

$$\begin{aligned}\sum_{t=1}^{T} \mathbb{E}_q\{\log p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t)\} &= -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log B_{p_0} - \frac{T}{2}\bar{\ell}_\sigma - \frac{1}{2}\sum_{t=1}^{T} \bar{\ell}_{v,t} \\ &\quad - \frac{\kappa}{2B_{p_0}} \sum_{t=1}^{T} \left\{ m_{1/v,t}\overline{r_t^2} - 2A_{p_0}\bar{r}_t + A_{p_0}^2 \bar{v}_t \right\}.\end{aligned} \tag{S46}$$

The entropy terms are given by (S159) for $q(\sigma)$ and by (S160) for the $q(v_t)$ factors.

S4. Dynamic exDQLM under the exAL likelihood

S4.1. Model hierarchy and posterior target

The dynamic exDQLM replaces the AL observation in (S27) with the exAL augmentation

$$v_t \mid \sigma \sim \mathrm{Exp}(\mathrm{rate} = 1/\sigma), \qquad s_t \sim \mathcal{N}^+(0,1), \tag{S47}$$

$$y_t \mid \boldsymbol{\theta}_t, \sigma, \gamma, v_t, s_t \sim \mathcal{N}\left(\mathbf{F}_t^\top \boldsymbol{\theta}_t + C_\gamma \sigma |\gamma| s_t + A_\gamma v_t,\ \sigma B_\gamma v_t\right), \tag{S48}$$

with the same state evolution as in Section S3. The complete posterior target is

$$\begin{aligned}p(\boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma \mid \boldsymbol{y}) \propto p(\boldsymbol{\theta}_0) \prod_{t=1}^{T} p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1}) \prod_{t=1}^{T} p(y_t \mid \boldsymbol{\theta}_t, v_t, s_t, \sigma, \gamma) \\ \times \prod_{t=1}^{T} p(v_t \mid \sigma) \prod_{t=1}^{T} p(s_t)\, p(\sigma) \pi_\gamma(\gamma).\end{aligned} \tag{S49}$$

Conditional on $(\boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma)$,

$$\tilde{y}_t = y_t - A_\gamma v_t - C_\gamma \sigma |\gamma| s_t, \qquad R_t^y = \sigma B_\gamma v_t, \tag{S50}$$

This is the generic Gaussian state update in Section S2.2 with $d_t = A_\gamma v_t + C_\gamma \sigma |\gamma| s_t$ and $Q_t = \sigma B_\gamma v_t$.

S4.2. MCMC updates

The default package path samples σ from its exact conditional given γ and then updates γ by bounded univariate slice sampling. Joint random-walk alternatives are also implemented, but the following target is the default exact-kernel path.

Latent v_t. Let $r_t = y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t - C_\gamma \sigma |\gamma| s_t$. Then

$$v_t \mid \cdot \sim \mathrm{GIG}\left(\frac{1}{2},\ \frac{r_t^2}{\sigma B_\gamma},\ \frac{A_\gamma^2}{\sigma B_\gamma} + \frac{2}{\sigma}\right). \tag{S51}$$

Latent s_t. Let $y_t^\circ = y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t$ and $D_\gamma = C_\gamma \sigma |\gamma|$. Then

$$s_t \mid \cdot \sim \mathcal{N}^+(\mu_{s,t}, V_{s,t}), \qquad V_{s,t} = \left(1 + \frac{D_\gamma^2}{\sigma B_\gamma v_t}\right)^{-1}, \qquad \mu_{s,t} = V_{s,t} \frac{D_\gamma y_t^\circ}{\sigma B_\gamma v_t}. \tag{S52}$$

State path. Given $(\boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma)$, sample $\boldsymbol{\theta}_{0:T}$ by the FFBS recursion in Section S2.2, using the exDQLM MCMC row of Table S3.

Scale. For fixed γ, let $y_t^\circ = y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t$ and $c_{\gamma,t} = C_\gamma |\gamma| s_t$. Then

$$\sigma \mid \cdot \sim \mathrm{GIG}(\lambda_\sigma, \chi_\sigma, \psi_\sigma), \tag{S53}$$

where

$$\lambda_\sigma = -\left(a_\sigma + \frac{3T}{2}\right), \tag{S54}$$

$$\chi_\sigma = 2b_\sigma + 2\sum_{t=1}^{T} v_t + \sum_{t=1}^{T} \frac{(y_t^\circ)^2}{B_\gamma v_t}, \tag{S55}$$

$$\psi_\sigma = \sum_{t=1}^{T} \frac{c_{\gamma,t}^2}{B_\gamma v_t}. \tag{S56}$$

Skewness. The conditional kernel for γ is

$$\pi(\gamma \mid \cdot) \propto \pi_\gamma(\gamma) \prod_{t=1}^{T} (\sigma B_\gamma v_t)^{-1/2} \exp\left[-\sum_{t=1}^{T} \frac{\{y_t - \mathbf{F}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t - C_\gamma \sigma |\gamma| s_t\}^2}{2\sigma B_\gamma v_t}\right], \tag{S57}$$

for $\gamma \in (L, U)$. The default sampler applies the bounded slice update in Section S9.2 directly to this kernel.

Algorithm S3. Dynamic exDQLM MCMC.

The default (σ, γ) update consists of an exact GIG update for σ conditional on γ, followed by a bounded univariate slice update for γ. Joint random-walk alternatives are implemented, but they are not the default kernel summarized here.

Input: observations $\boldsymbol{y}$, model matrices $(\mathbf{F}_t, \mathbf{G}_t, \mathbf{W}_t)$, initial state prior, exAL prior hyperparameters, admissible interval (L, U) for γ, and MCMC controls N_{burn} and N_{mcmc}.

Output: retained posterior draws for $(\boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma)$ and state summaries.

(1) Initialize $\boldsymbol{\theta}_{0:T}$, $\boldsymbol{v}$, $\boldsymbol{s}$, σ, and γ.

(2) For $m = 1, \ldots, N_{\text{burn}} + N_{\text{mcmc}}$:

(a) sample v_t from (S51), $t = 1, \dots, T$;
(b) sample s_t from (S52), $t = 1, \dots, T$;
(c) sample $\boldsymbol{\theta}_{0:T}$ by FFBS using Section S2.2 and the exDQLM MCMC row of Table S3;
(d) sample σ from (S53);
(e) sample γ on (L, U) with the bounded slice sampler in Section S9.2, targeting (S57);
(f) if $m > N_{\text{burn}}$, retain the current draw.

(3) Return the retained draws and state summaries.

Setting $\gamma = 0$ drops the s_t block and reduces the sampler to the AL/DQLM kernel in Algorithm S1.

S4.3. Laplace–delta VB approximation

The mean-field family is

$$q(\boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma) = q(\boldsymbol{\theta}_{0:T}) \prod_{t=1}^{T} q(v_t) \prod_{t=1}^{T} q(s_t)\, q(\sigma, \gamma). \tag{S58}$$

The conjugate factors are updated conditionally on moments from $q(\sigma, \gamma)$. Define

$$\kappa_1 = \mathbb{E}\{1/(\sigma B_\gamma)\}, \qquad \kappa_2 = \mathbb{E}\{A_\gamma/(\sigma B_\gamma)\}, \qquad \kappa_3 = \mathbb{E}\{A_\gamma^2/(\sigma B_\gamma)\}, \tag{S59}$$

$$\kappa_4 = \mathbb{E}\{C_\gamma|\gamma|/B_\gamma\}, \qquad \kappa_5 = \mathbb{E}\{\sigma C_\gamma^2 \gamma^2/B_\gamma\}, \qquad \kappa_6 = \mathbb{E}\{A_\gamma C_\gamma|\gamma|/B_\gamma\}. \tag{S60}$$

Let $m_{1/v,t} = \mathbb{E}(v_t^{-1})$, $\bar{s}_t = \mathbb{E}(s_t)$, and $\overline{s_t^2} = \mathbb{E}(s_t^2)$. The state factor is Gaussian and is updated by the deterministic smoother in Section S2.2 with observation precision

$$\phi_t = \kappa_1 m_{1/v,t} \tag{S61}$$

and pseudo-response

$$y_t^{\text{VB}} = \frac{y_t \phi_t - \kappa_2 - \kappa_4 \bar{s}_t m_{1/v,t}}{\phi_t}. \tag{S62}$$

Equivalently, in the original-data form (S12), the offset is $d_t = \{\kappa_2 + \kappa_4 \bar{s}_t m_{1/v,t}\}/\phi_t$ and the variance is $Q_t = \phi_t^{-1}$. To see this, let $\eta_t = \mathbf{F}_t^\top \boldsymbol{\theta}_t$ and let $\mathbb{E}_{-\theta}$ denote expectation over all variational factors except the state factor. The terms in the expected quadratic loss that depend on η_t are

$$\begin{aligned} &\mathbb{E}_{-\theta}\left[\frac{\{y_t - \eta_t - A_\gamma v_t - C_\gamma \sigma|\gamma| s_t\}^2}{\sigma B_\gamma v_t}\right] \\ &\quad = \phi_t (y_t - \eta_t)^2 - 2\{\kappa_2 + \kappa_4 \bar{s}_t m_{1/v,t}\}(y_t - \eta_t) + \text{constant}, \end{aligned} \tag{S63}$$

so completing the square gives (S62) and (S61). If $\eta_t = \mathbf{F}_t^\top \boldsymbol{\theta}_t$, write $m_{\eta,t} = \mathbb{E}(\eta_t)$, $S_{\eta,t} = \text{Var}(\eta_t)$, $m_{\Delta,t} = y_t - m_{\eta,t}$, and $S_{\Delta,t} = m_{\Delta,t}^2 + S_{\eta,t}$. Then

$$q(v_t) = \text{GIG}\left(\frac{1}{2},\ \kappa_1 S_{\Delta,t} - 2\kappa_4 m_{\Delta,t} \bar{s}_t + \kappa_5 \overline{s_t^2},\ \kappa_3 + 2\mathbb{E}(\sigma^{-1})\right), \tag{S64}$$

$$q(s_t) = \mathcal{N}^+(\mu_{s,t}^{\text{VB}}, V_{s,t}^{\text{VB}}), \tag{S65}$$

$$V_{s,t}^{\text{VB}} = \left(1 + \kappa_5 m_{1/v,t}\right)^{-1}, \tag{S66}$$

$$\mu_{s,t}^{\text{VB}} = V_{s,t}^{\text{VB}} \left(\kappa_4 m_{\Delta,t} m_{1/v,t} - \kappa_6\right). \tag{S67}$$

The nonconjugate factor $q(\sigma, \gamma)$ is approximated by a Laplace–delta step on transformed coordinates

$$\ell_\sigma = \log \sigma, \qquad z_\gamma = \operatorname{logit}\left(\frac{\gamma - L}{U - L}\right). \tag{S68}$$

The transformed objective is the expected complete log-joint plus the Jacobian. Here $\mathbb{E}_{-(\sigma,\gamma)}$ denotes expectation with respect to all variational factors except $q(\sigma, \gamma)$:

$$\begin{aligned} h_z(\ell_\sigma, z_\gamma) = \mathbb{E}_{-(\sigma,\gamma)}\{\log p(\boldsymbol{y}, \boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma)\} \\ + \ell_\sigma + \log(U - L) + \log r + \log(1 - r), \qquad r = \operatorname{expit}(z_\gamma). \end{aligned} \tag{S69}$$

The fitted object returns the transformed mode, the approximate covariance matrix, delta-method moments, and ELBO components.

Algorithm S4. Dynamic exDQLM LDVB.

Input: observations $\boldsymbol{y}$, model matrices $(\mathbf{F}_t, \mathbf{G}_t, \mathbf{W}_t)$, initial state prior, exAL prior hyper-parameters, admissible interval (L, U) for γ, convergence tolerance, and optional posterior-sampling control `n.samp`.

Output: fitted variational factors for $(\boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma)$, convergence diagnostics, ELBO values when requested, and optional approximate posterior draws.

(1) Initialize variational parameters for $q(\boldsymbol{\theta}_{0:T})$, $q(v_t)$, $q(s_t)$, and $q(\sigma, \gamma)$.

(2) Repeat until the convergence criterion is met:

  (a) update $q(\boldsymbol{\theta}_{0:T})$ by the deterministic smoother in Section S2.2, using the exDQLM LDVB row of Table S3;

  (b) update $q(v_t)$ and $q(s_t)$ using (S67);

  (c) update $q(\sigma, \gamma)$ on the transformed coordinates (ℓ_σ, z_γ) in (S68) by the Laplace–delta approximation to (S69);

  (d) monitor moment changes and, when ELBO monitoring is enabled, $\mathcal{L}(q)$.

(3) Draw approximate posterior samples from the fitted variational factors when requested by `n.samp`.

(4) Return the fitted factors, diagnostics, and optional draws.

In the AL special case, the nonconjugate (σ, γ) block is replaced by the inverse-gamma $q(\sigma)$ update in Algorithm S2. This exAL path is the default in `exdqlmMCMC()` and `exdqlmLDVB()` when `dqlm.ind = FALSE`.

ELBO. The LDVB ELBO has the block decomposition

$$\mathcal{L}(q) = \mathcal{L}_{\text{like}} + \mathcal{L}_\theta + \mathcal{L}_v + \mathcal{L}_s + \mathcal{L}_{\sigma\gamma} + \sum_{t=1}^{T} H\{q(v_t)\} + \sum_{t=1}^{T} H\{q(s_t)\} + H\{q(\sigma, \gamma)\}. \tag{S70}$$

The state block is

$$\mathcal{L}_\theta = \mathbb{E}_q\left\{\log p(\boldsymbol{\theta}_0) + \sum_{t=1}^{T} \log p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1})\right\} + H\{q(\boldsymbol{\theta}_{0:T})\}. \tag{S71}$$

As in the AL case, this Gaussian state-space contribution can be evaluated by the auxiliary Kalman normalizing-constant identity for the current pseudo-response and variance.

Let $\bar{\ell}_\sigma = \mathbb{E}(\log\sigma)$, $\bar{\ell}_B = \mathbb{E}(\log B_\gamma)$, $\bar{\ell}_\pi = \mathbb{E}\{\log\pi_\gamma(\gamma)\}$, $m_{v,t} = \mathbb{E}(v_t)$, $m_{1/v,t} = \mathbb{E}(v_t^{-1})$, $m_{\log v,t} = \mathbb{E}(\log v_t)$, $\bar{s}_t = \mathbb{E}(s_t)$, and $\overline{s_t^2} = \mathbb{E}(s_t^2)$. Define

$$\begin{aligned}\mathcal{Q}_t = \kappa_1 S_{\Delta,t} m_{1/v,t} - 2\kappa_2 m_{\Delta,t} + \kappa_3 m_{v,t} - 2\kappa_4 m_{\Delta,t}\bar{s}_t m_{1/v,t} \\ + \kappa_5 \overline{s_t^2} m_{1/v,t} + 2\kappa_6 \bar{s}_t.\end{aligned} \tag{S72}$$

Then the likelihood, latent-prior, and nonconjugate-prior blocks are

$$\mathcal{L}_{\text{like}} = -\frac{T}{2}\log(2\pi) - \frac{T}{2}(\bar{\ell}_\sigma + \bar{\ell}_B) - \frac{1}{2}\sum_{t=1}^{T} m_{\log v,t} - \frac{1}{2}\sum_{t=1}^{T}\mathcal{Q}_t, \tag{S73}$$

$$\mathcal{L}_v = -T\bar{\ell}_\sigma - \mathbb{E}(\sigma^{-1})\sum_{t=1}^{T} m_{v,t}, \tag{S74}$$

$$\mathcal{L}_s = T\log 2 - \frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\overline{s_t^2}, \tag{S75}$$

$$\mathcal{L}_{\sigma\gamma} = a_\sigma \log b_\sigma - \log\Gamma(a_\sigma) - (a_\sigma + 1)\bar{\ell}_\sigma - b_\sigma\mathbb{E}(\sigma^{-1}) + \bar{\ell}_\pi. \tag{S76}$$

The GIG and truncated-normal entropy terms are given by (S160) and (S161). The (σ,γ) entropy and the smooth expectations in the κ_j, $\bar{\ell}_B$, $\bar{\ell}_\pi$, and $\mathbb{E}(\sigma^{-1})$ terms are evaluated under the Laplace–delta approximation in (S163) and (S166).

S5. Transfer-function exDQLM by state augmentation

Transfer-function inference uses the same AL/exAL likelihood and latent-variable updates as the dynamic models above. The only change is the state-space construction. Let $x_t \in \mathbb{R}^k$ be the input vector, ζ_t the transfer state, and $\psi_t \in \mathbb{R}^k$ the input-effect state. For fixed transfer rate λ, the augmented model uses

$$\widetilde{\mathbf{F}}_t^\top = (\mathbf{F}_t^\top,\ 1,\ 0_k^\top), \qquad \widetilde{\boldsymbol{\theta}}_t^\top = (\boldsymbol{\theta}_t^\top,\ \zeta_t,\ \psi_t^\top), \tag{S77}$$

$$\widetilde{\mathbf{G}}_t = \text{blockdiag}\left\{\mathbf{G}_t, \begin{pmatrix}\lambda & x_t^\top \\ 0_k & I_k\end{pmatrix}\right\}, \qquad \widetilde{\mathbf{W}}_t = \text{blockdiag}\{\mathbf{W}_t^\theta, \omega_t, \mathbf{W}_t^\psi\}. \tag{S78}$$

The prior for $(\zeta_0, \psi_0^\top)^\top$ is $\mathcal{N}(m_0^{\text{tf}}, C_0^{\text{tf}})$ and is appended to the base state prior. Conditional on λ, the augmented model has the same posterior form as the base exDQLM with state $\widetilde{\boldsymbol{\theta}}_t$. Consequently, the Gaussian state update in Section S2.2, together with Algorithms S3 and S4, applies after replacing $(\mathbf{F}_t, \mathbf{G}_t, \mathbf{W}_t, \boldsymbol{\theta}_t)$ by $(\widetilde{\mathbf{F}}_t, \widetilde{\mathbf{G}}_t, \widetilde{\mathbf{W}}_t, \widetilde{\boldsymbol{\theta}}_t)$. The current transfer-fitting routines treat λ as fixed by the user or by an external grid search; they do not sample or optimize λ internally. The package inputs `X`, `lam`, `tf.df`, `tf.m0`, and `tf.C0` correspond to x_t, λ, the transfer discount factors, m_0^{tf}, and C_0^{tf}; `median.kt` is returned as a summary of the fitted transfer response.

Algorithm S5. Transfer-function state augmentation and fitting.

Input: base dynamic model, transfer covariates x_t, fixed transfer rate λ, transfer discount factors, and transfer-state prior $(m_0^{\text{tf}}, C_0^{\text{tf}})$.

Output: fitted base-state and transfer-state summaries, including the transfer response summary `median.kt`.

(1) Construct $\widetilde{\mathbf{F}}_t$, $\widetilde{\mathbf{G}}_t$, $\widetilde{\mathbf{W}}_t$, and $\widetilde{\boldsymbol{\theta}}_t$ from (S77)–(S78).

(2) Set the augmented initial prior.

(3) Run the selected fitting routine on the augmented state-space system: MCMC uses Algorithm S3 and LDVB uses Algorithm S4.

(4) Return summaries for both the base state and the transfer states.

The transfer-function extension changes only the state-space matrices; the observation likelihood and latent-variable updates are unchanged.

S6. Static AL and exAL regression with Gaussian priors

S6.1. Model hierarchy and posterior target

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ have row vectors $\boldsymbol{x}_i^\top$, and let $\boldsymbol{\beta} \in \mathbb{R}^p$. The ridge/Gaussian coefficient prior is

$$\boldsymbol{\beta} \sim \mathcal{N}(\boldsymbol{b}_{\beta,0}, \mathbf{V}_{\beta,0}), \qquad \boldsymbol{b}_{\beta,0} \in \mathbb{R}^p, \quad \mathbf{V}_{\beta,0} \in \mathbb{R}^{p\times p}. \tag{S79}$$

In the package interface, the same Gaussian update is used for `beta_prior = "ridge"`. When `beta_prior = "rhs_ns"`, the diagonal prior precision in this update is replaced by the `rhs_ns` expected precision described in Section S7. Thus this supplement covers the Gaussian/ridge prior and the conjugate `rhs_ns` prior, both for MCMC and LDVB. Under the exAL likelihood,

$$v_i \mid \sigma \sim \mathrm{Exp}(\mathrm{rate} = 1/\sigma), \qquad s_i \sim \mathcal{N}^+(0,1), \tag{S80}$$

$$y_i \mid \boldsymbol{\beta}, \sigma, \gamma, v_i, s_i \sim \mathcal{N}\left(\boldsymbol{x}_i^\top\boldsymbol{\beta} + C_\gamma\sigma|\gamma|s_i + A_\gamma v_i,\ \sigma B_\gamma v_i\right). \tag{S81}$$

The static AL model is obtained by setting $\gamma = 0$, removing s_i, and using A_{p_0}, B_{p_0}. The exAL posterior target is

$$p(\boldsymbol{\beta}, \sigma, \gamma, \boldsymbol{v}, \boldsymbol{s} \mid \boldsymbol{y}) \propto p(\boldsymbol{\beta})p(\sigma)\pi_\gamma(\gamma)\prod_{i=1}^n p(y_i \mid \boldsymbol{\beta}, \sigma, \gamma, v_i, s_i)p(v_i \mid \sigma)p(s_i). \tag{S82}$$

Unlike the dynamic models in Sections S3–S5, the static regression update does not use Kalman filtering, FFBS, or RTS smoothing. The coefficient block is a Gaussian weighted-regression update.

S6.2. MCMC updates

For fixed $(\sigma, \gamma, \boldsymbol{v}, \boldsymbol{s})$, define

$$y_i^\star = y_i - C_\gamma\sigma|\gamma|s_i - A_\gamma v_i, \qquad w_i = (\sigma B_\gamma v_i)^{-1}. \tag{S83}$$

Let $\mathbf{W} = \operatorname{diag}(w_1, \ldots, w_n)$ and $\boldsymbol{y}^\star = (y_1^\star, \ldots, y_n^\star)^\top$. Then

$$\boldsymbol{\Sigma}_{\beta|\cdot} = \left(\mathbf{V}_{\beta,0}^{-1} + \mathbf{X}^\top \mathbf{W} \mathbf{X}\right)^{-1}, \tag{S84}$$

$$\boldsymbol{m}_{\beta|\cdot} = \boldsymbol{\Sigma}_{\beta|\cdot} \left(\mathbf{V}_{\beta,0}^{-1} \boldsymbol{b}_{\beta,0} + \mathbf{X}^\top \mathbf{W} \boldsymbol{y}^\star\right), \tag{S85}$$

$$\boldsymbol{\beta} \mid \cdot \sim \mathcal{N}(\boldsymbol{m}_{\beta|\cdot}, \boldsymbol{\Sigma}_{\beta|\cdot}). \tag{S86}$$

Let $D_\gamma = C_\gamma |\gamma|$. The remaining exAL full conditionals are

$$s_i \mid \cdot \sim \mathcal{N}^+(\mu_{s,i}, V_{s,i}), \tag{S87}$$

$$V_{s,i} = \left(1 + \frac{\sigma D_\gamma^2}{B_\gamma v_i}\right)^{-1}, \tag{S88}$$

$$\mu_{s,i} = V_{s,i} \frac{D_\gamma (y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i)}{B_\gamma v_i}, \tag{S89}$$

and, with $r_i = y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} - \sigma D_\gamma s_i$,

$$v_i \mid \cdot \sim \operatorname{GIG}\left(\frac{1}{2}, \ \frac{r_i^2}{\sigma B_\gamma}, \ \frac{A_\gamma^2}{\sigma B_\gamma} + \frac{2}{\sigma}\right). \tag{S90}$$

For fixed γ, define $t_i = y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i$. Then

$$\sigma \mid \cdot \sim \operatorname{GIG}\left(-a_\sigma - \frac{3n}{2}, \ 2b_\sigma + 2\sum_{i=1}^n v_i + \sum_{i=1}^n \frac{t_i^2}{B_\gamma v_i}, \ \sum_{i=1}^n \frac{D_\gamma^2 s_i^2}{B_\gamma v_i}\right). \tag{S91}$$

The nonconjugate conditional kernel for γ is

$$\pi(\gamma \mid \cdot) \propto \pi_\gamma(\gamma) \, [B_\gamma]^{-n/2} \exp\left\{\sum_{i=1}^n \frac{D_\gamma s_i t_i}{B_\gamma v_i} - \frac{1}{2\sigma} \sum_{i=1}^n \frac{t_i^2}{B_\gamma v_i} - \frac{\sigma}{2} \sum_{i=1}^n \frac{D_\gamma^2 s_i^2}{B_\gamma v_i}\right\}, \tag{S92}$$

on (L, U). The package default updates this one-dimensional target by the bounded slice sampler in Section S9.2.

For the AL special case, the Gibbs updates reduce to

$$\boldsymbol{\beta} \mid \cdot \sim \mathcal{N}(\boldsymbol{m}_{\beta|\cdot}, \boldsymbol{\Sigma}_{\beta|\cdot}), \tag{S93}$$

$$\sigma \mid \cdot \sim \operatorname{IG}\left(a_\sigma + \frac{3n}{2}, \ b_\sigma + \sum_{i=1}^n v_i + \sum_{i=1}^n \frac{(y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} - A_{p_0} v_i)^2}{2B_{p_0} v_i}\right), \tag{S94}$$

$$v_i \mid \cdot \sim \operatorname{GIG}\left(\frac{1}{2}, \ \frac{(y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta})^2}{\sigma B_{p_0}}, \ \frac{A_{p_0}^2}{\sigma B_{p_0}} + \frac{2}{\sigma}\right). \tag{S95}$$

Algorithm S6. Static AL/exAL MCMC.

Input: response vector $\boldsymbol{y}$, design matrix $\mathbf{X}$, target quantile p_0, coefficient prior specification, AL/exAL prior hyperparameters, and MCMC controls N_{burn} and N_{mcmc}.

Output: retained posterior draws for regression coefficients, latent variables, scale parameters, and shrinkage parameters when `beta_prior = "rhs_ns"`.

(1) Initialize $\boldsymbol{\beta}$, $\boldsymbol{v}$, $\boldsymbol{s}$, σ, and γ; omit $\boldsymbol{s}$ and fix $\gamma = 0$ for the AL path.

(2) For $m = 1, \ldots, N_{\text{burn}} + N_{\text{mcmc}}$:

   (a) sample $\boldsymbol{\beta}$ from (S86);

   (b) sample v_i from (S90), $i = 1, \ldots, n$, with the AL simplification in (S95);

   (c) for exAL, sample s_i from (S89), $i = 1, \ldots, n$;

   (d) sample σ from (S91), with the AL simplification in (S94);

   (e) for exAL, sample γ on (L, U) with the bounded slice sampler, targeting (S92);

   (f) if `beta_prior = "rhs_ns"`, update the `rhs_ns` scale block in (S126);

   (g) if $m > N_{\text{burn}}$, retain the current draw.

(3) Return the retained posterior draws.

S6.3. Laplace–delta VB updates and ELBO

The static exAL variational family is

$$q(\boldsymbol{\beta}, \boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma) = q(\boldsymbol{\beta}) \prod_{i=1}^{n} q(v_i) \prod_{i=1}^{n} q(s_i) q(\sigma, \gamma). \tag{S96}$$

The coefficient factor is Gaussian. With the same moment definitions as in (S60),

$$\omega_i = \kappa_1 \mathbb{E}(v_i^{-1}), \tag{S97}$$

$$\eta_i = y_i \kappa_1 \mathbb{E}(v_i^{-1}) - \kappa_4 \mathbb{E}(v_i^{-1}) \mathbb{E}(s_i) - \kappa_2, \tag{S98}$$

$$\boldsymbol{\Sigma}_\beta = \left(\mathbf{V}_{\beta,0}^{-1} + \mathbf{X}^\top \operatorname{diag}(\omega_1, \ldots, \omega_n) \mathbf{X} \right)^{-1}, \tag{S99}$$

$$\boldsymbol{m}_\beta = \boldsymbol{\Sigma}_\beta \left(\mathbf{V}_{\beta,0}^{-1} \boldsymbol{b}_{\beta,0} + \mathbf{X}^\top \boldsymbol{\eta} \right), \tag{S100}$$

where $\boldsymbol{\eta} = (\eta_1, \ldots, \eta_n)^\top$. The $q(v_i)$ and $q(s_i)$ updates are also closed form conditional on moments of $q(\sigma, \gamma)$. Let

$$\bar{r}_i = y_i - \boldsymbol{x}_i^\top \boldsymbol{m}_\beta, \qquad \overline{r_i^2} = \bar{r}_i^2 + \boldsymbol{x}_i^\top \boldsymbol{\Sigma}_\beta \boldsymbol{x}_i. \tag{S101}$$

Then

$$q(v_i) = \text{GIG}\left(\frac{1}{2},\ \kappa_1 \overline{r_i^2} - 2\kappa_4 \bar{r}_i \mathbb{E}(s_i) + \kappa_5 \mathbb{E}(s_i^2),\ \kappa_3 + 2\mathbb{E}(\sigma^{-1}) \right), \tag{S102}$$

$$q(s_i) = \mathcal{N}^+(\mu_{s,i}^{\text{VB}}, V_{s,i}^{\text{VB}}), \tag{S103}$$

$$V_{s,i}^{\text{VB}} = \left(1 + \kappa_5 \mathbb{E}(v_i^{-1}) \right)^{-1}, \tag{S104}$$

$$\mu_{s,i}^{\text{VB}} = V_{s,i}^{\text{VB}} \left\{ \kappa_4 \bar{r}_i \mathbb{E}(v_i^{-1}) - \kappa_6 \right\}. \tag{S105}$$

The nonconjugate factor $q(\sigma, \gamma)$ is proportional to

$$q(\sigma, \gamma) \propto \pi_\gamma(\gamma) [B_\gamma]^{-n/2} \sigma^{-(a_\sigma + 1 + 3n/2)} \exp\left\{ -\frac{1}{2} \left(\frac{\bar{c}(\gamma)}{\sigma} + \bar{d}(\gamma)\sigma \right) + \bar{e}(\gamma) \right\}, \tag{S106}$$

where

$$\bar{c}(\gamma) = 2b_\sigma + 2\sum_{i=1}^{n}\mathbb{E}(v_i) + \sum_{i=1}^{n}\frac{\mathbb{E}\{(y_i - \boldsymbol{x}_i^\top\boldsymbol{\beta} - A_\gamma v_i)^2/v_i\}}{B_\gamma}, \tag{S107}$$

$$\bar{d}(\gamma) = \sum_{i=1}^{n}\frac{C_\gamma^2\gamma^2}{B_\gamma}\mathbb{E}\{s_i^2/v_i\}, \tag{S108}$$

$$\bar{e}(\gamma) = \sum_{i=1}^{n}\frac{C_\gamma|\gamma|}{B_\gamma}\mathbb{E}\{s_i(y_i - \boldsymbol{x}_i^\top\boldsymbol{\beta} - A_\gamma v_i)/v_i\}. \tag{S109}$$

This factor is approximated by the same Laplace–delta method as in the dynamic exDQLM. The AL path sets $\gamma = 0$ and uses $q(\boldsymbol{\beta})q(\sigma)\prod_i q(v_i)$ with the same closed-form updates as Section S3, replacing state smoothing moments by $\mathbb{E}\{(y_i - \boldsymbol{x}_i^\top\boldsymbol{\beta})^2\}$.

Algorithm S7. Static AL/exAL LDVB.

Input: response vector $\boldsymbol{y}$, design matrix $\mathbf{X}$, target quantile p_0, coefficient prior specification, AL/exAL prior hyperparameters, convergence tolerance, and optional posterior-sampling control `n.samp`.

Output: fitted variational factors, convergence diagnostics, ELBO values, and optional approximate posterior draws.

(1) Initialize $q(\boldsymbol{\beta})$, the latent-variable factors, the scale factors, and the (σ, γ) block.

(2) Repeat until the convergence criterion is met:

   (a) update $q(\boldsymbol{\beta})$ using (S100);

   (b) update $q(v_i)$ and, for exAL, $q(s_i)$ using (S105);

   (c) update $q(\sigma)$ for AL or $q(\sigma, \gamma)$ for exAL using the Laplace–delta approximation to (S106);

   (d) if `beta_prior = "rhs_ns"`, update the `rhs_ns` factors in (S134);

   (e) monitor moment changes and $\mathcal{L}(q)$.

(3) Draw approximate posterior samples when requested by `n.samp`.

(4) Return the fitted factors, diagnostics, and optional draws.

These static updates are implemented in `exalStaticMCMC()` and `exalStaticLDVB()`, with `beta_prior` selecting the Gaussian/ridge or `rhs_ns` coefficient prior.

The static exAL ELBO is computed from the same blocks as the dynamic exDQLM, with the Gaussian state factor replaced by the Gaussian coefficient factor. Let $\bar{\ell}_\sigma = \mathbb{E}(\log\sigma)$, $\bar{\ell}_B = \mathbb{E}(\log B_\gamma)$, $\bar{\ell}_\pi = \mathbb{E}\{\log\pi_\gamma(\gamma)\}$, $m_{v,i} = \mathbb{E}(v_i)$, $m_{1/v,i} = \mathbb{E}(v_i^{-1})$, $m_{\log v,i} = \mathbb{E}(\log v_i)$, $\bar{s}_i = \mathbb{E}(s_i)$, and $\overline{s_i^2} = \mathbb{E}(s_i^2)$. Also define

$$t_i = y_i - \boldsymbol{x}_i^\top\boldsymbol{m}_\beta, \qquad q_i = \boldsymbol{x}_i^\top\boldsymbol{\Sigma}_\beta\boldsymbol{x}_i. \tag{S110}$$

Then

$$\mathcal{L}(q) = \mathcal{L}_{\text{like}} + \mathcal{L}_v + \mathcal{L}_s + \mathcal{L}_\beta + \mathcal{L}_{\sigma\gamma} + H\{q(\boldsymbol{\beta})\} + \sum_{i=1}^n H\{q(v_i)\}$$
$$+ \sum_{i=1}^n H\{q(s_i)\} + H\{q(\sigma, \gamma)\}, \tag{S111}$$

where

$$\begin{aligned}
\mathcal{L}_{\text{like}} &= -\frac{n}{2}\log(2\pi) - \frac{n}{2}(\bar{\ell}_B + \bar{\ell}_\sigma) - \frac{1}{2}\sum_{i=1}^n m_{\log v, i} \\
&\quad - \frac{1}{2}\sum_{i=1}^n \left\{\kappa_1 m_{1/v,i}(t_i^2 + q_i) - 2\kappa_2 t_i + \kappa_3 m_{v,i}\right\} \\
&\quad + \sum_{i=1}^n \left\{\kappa_4 \bar{s}_i m_{1/v,i} t_i - \kappa_6 \bar{s}_i - \frac{1}{2}\kappa_5 \overline{s_i^2} m_{1/v,i}\right\}, && \text{(S112)} \\
\mathcal{L}_v &= -n\bar{\ell}_\sigma - \mathbb{E}(\sigma^{-1})\sum_{i=1}^n m_{v,i}, && \text{(S113)} \\
\mathcal{L}_s &= n\log 2 - \frac{n}{2}\log(2\pi) - \frac{1}{2}\sum_{i=1}^n \overline{s_i^2}, && \text{(S114)} \\
\mathcal{L}_{\sigma\gamma} &= a_\sigma \log b_\sigma - \log\Gamma(a_\sigma) - (a_\sigma + 1)\bar{\ell}_\sigma - b_\sigma \mathbb{E}(\sigma^{-1}) + \bar{\ell}_\pi, && \text{(S115)} \\
\mathcal{L}_\beta &= \mathbb{E}_q\{\log p(\boldsymbol{\beta})\}. && \text{(S116)}
\end{aligned}$$

For the ridge prior, $\mathcal{L}_\beta$ is the Gaussian prior expectation. For the `rhs_ns` prior, it is replaced by the `rhs_ns` contribution in Section S7. The Gaussian, GIG, truncated-normal, and Laplace–delta entropy terms are given by (S157), (S160), (S161), and (S163), respectively. The AL path is recovered by setting $\gamma = 0$ and dropping the s_i and (σ, γ) Laplace–delta blocks; then $q(\sigma)$ is inverse-gamma and uses (S159).

S7. Static regression with the Nishimura–Suchard regularized horseshoe prior

This section focuses on the `rhs_ns` coefficient prior, the conditionally conjugate regularized horseshoe variant used for sparse static regression. Users only need this section when fitting static models with `beta_prior = "rhs_ns"`. The prior enters the static Gaussian update for $\boldsymbol{\beta}$ through a prior precision matrix. Let $\mathcal{A} \subseteq \{1, \ldots, p\}$ denote the active shrinkage set; by default an intercept, if present, is not shrunk. For $j \notin \mathcal{A}$ the package uses a fixed intercept precision. The direct `rhs` path is not developed here because its log-scale block is nonconjugate; the `rhs_ns` path is the conjugate sparse-prior formulation documented in this supplement.

S7.1. Nishimura–Suchard prior hierarchy

For $j \in \mathcal{A}$, the `rhs_ns` hierarchy is

$$\beta_j \mid \tau^2, \lambda_j^2, \zeta^2 \propto \mathcal{N}(\beta_j \mid 0, \tau^2\lambda_j^2)\mathcal{N}(\beta_j \mid 0, \zeta^2), \tag{S117}$$

$$\lambda_j^2 \mid \nu_j \sim \mathrm{IG}(1/2, 1/\nu_j), \qquad \nu_j \sim \mathrm{IG}(1/2, 1), \tag{S118}$$

$$\tau^2 \mid \xi \sim \mathrm{IG}(1/2, 1/\xi), \qquad \xi \sim \mathrm{IG}(1/2, 1/\tau_0^2), \tag{S119}$$

$$\zeta^2 \sim \mathrm{IG}(a_\zeta, b_\zeta), \tag{S120}$$

unless ζ^2 is fixed by the user. The implementation includes the scale-dependent Gaussian log-normalizing terms shown in the ELBO contribution below. Thus the proportionality in (S120) denotes the coefficient kernel, while the surrounding scale updates and ELBO use the corresponding normal-density contributions. The resulting prior precision contribution for active coefficients is

$$\omega_j^{\mathrm{prior}} = \frac{1}{\tau^2\lambda_j^2} + \frac{1}{\zeta^2}. \tag{S121}$$

Consequently, the $\boldsymbol{\beta}$ update in MCMC or LDVB is obtained from the static Gaussian update after replacing $\mathbf{V}_{\beta,0}^{-1}$ by a diagonal precision matrix with active entries $\omega_j^{\mathrm{prior}}$ or their variational expectations.

S7.2. Nishimura–Suchard MCMC updates

Conditioning on $\boldsymbol{\beta}$, the shrinkage block has closed-form inverse-gamma updates:

$$\lambda_j^2 \mid \cdot \sim \mathrm{IG}\left(1,\ \frac{1}{\nu_j} + \frac{\beta_j^2}{2\tau^2}\right), \tag{S122}$$

$$\nu_j \mid \cdot \sim \mathrm{IG}\left(1,\ 1 + \frac{1}{\lambda_j^2}\right), \tag{S123}$$

$$\tau^2 \mid \cdot \sim \mathrm{IG}\left(\frac{|\mathcal{A}|+1}{2},\ \frac{1}{\xi} + \frac{1}{2}\sum_{j\in\mathcal{A}}\frac{\beta_j^2}{\lambda_j^2}\right), \tag{S124}$$

$$\xi \mid \cdot \sim \mathrm{IG}\left(1,\ \frac{1}{\tau_0^2} + \frac{1}{\tau^2}\right), \tag{S125}$$

$$\zeta^2 \mid \cdot \sim \mathrm{IG}\left(a_\zeta + \frac{|\mathcal{A}|}{2},\ b_\zeta + \frac{1}{2}\sum_{j\in\mathcal{A}}\beta_j^2\right), \tag{S126}$$

with the last line omitted when ζ^2 is fixed. These equations are the conditional targets for the `rhs_ns` scale block. The implementation may temporarily freeze or schedule the global-scale update during configured warmup iterations for numerical stability; this changes only the update schedule, not the conditional target displayed above.

S7.3. Nishimura–Suchard variational updates and ELBO contribution

The package uses a mean-field approximation over the `rhs_ns` scale block. For $j \in \mathcal{A}$,

$$q(\lambda_j^2) = \mathrm{IG}(a_{\lambda j}, b_{\lambda j}), \qquad q(\nu_j) = \mathrm{IG}(a_{\nu j}, b_{\nu j}), \tag{S127}$$
$$q(\tau^2) = \mathrm{IG}(a_\tau, b_\tau), \qquad q(\xi) = \mathrm{IG}(a_\xi, b_\xi), \tag{S128}$$
$$q(\zeta^2) = \mathrm{IG}(a_\zeta^\star, b_\zeta^\star), \tag{S129}$$

with $a_{\lambda j} = a_{\nu j} = a_\xi = 1$ and $a_\tau = (|\mathcal{A}| + 1)/2$. Let $\bar{\beta}_j^2 = \mathbb{E}_q(\beta_j^2)$. The coordinate updates are

$$b_{\lambda j} = \frac{1}{2}\bar{\beta}_j^2\, \mathbb{E}\{(\tau^2)^{-1}\} + \mathbb{E}(\nu_j^{-1}), \tag{S130}$$
$$b_{\nu j} = 1 + \mathbb{E}\{(\lambda_j^2)^{-1}\}, \tag{S131}$$
$$b_\tau = \frac{1}{2}\sum_{j\in\mathcal{A}} \bar{\beta}_j^2 \mathbb{E}\{(\lambda_j^2)^{-1}\} + \mathbb{E}(\xi^{-1}), \tag{S132}$$
$$b_\xi = \frac{1}{\tau_0^2} + \mathbb{E}\{(\tau^2)^{-1}\}, \tag{S133}$$
$$a_\zeta^\star = a_\zeta + \frac{|\mathcal{A}|}{2}, \qquad b_\zeta^\star = b_\zeta + \frac{1}{2}\sum_{j\in\mathcal{A}} \bar{\beta}_j^2. \tag{S134}$$

The third line is shorthand for $b_\tau = \frac{1}{2}\sum_{j\in\mathcal{A}} \bar{\beta}_j^2 \mathbb{E}\{(\lambda_j^2)^{-1}\} + \mathbb{E}(\xi^{-1})$, where the expectation of $(\lambda_j^2)^{-1}$ is component-specific inside the summation. As in MCMC, optional warmup scheduling for the τ^2, ξ block affects when the coordinate is updated, not the coordinate target itself.

The following term-by-term ELBO contribution matches the package's `rhs_ns` scale block. For $x \sim \mathrm{IG}(a, b)$ under the rate parameterization used above, define

$$m_{\log x}(a, b) = \log b - \psi(a), \qquad m_{x^{-1}}(a, b) = a/b, \tag{S135}$$
$$h_{\mathrm{IG}}(a, b) = a + \log b + \log\Gamma(a) - (a + 1)\psi(a), \tag{S136}$$

where $\psi(\cdot)$ is the digamma function and h_{IG} is the entropy of an inverse-gamma variational factor. Let

$$\ell_{\lambda j} = m_{\log x}(a_{\lambda j}, b_{\lambda j}), \qquad r_{\lambda j} = m_{x^{-1}}(a_{\lambda j}, b_{\lambda j}), \tag{S137}$$
$$\ell_{\nu j} = m_{\log x}(a_{\nu j}, b_{\nu j}), \qquad r_{\nu j} = m_{x^{-1}}(a_{\nu j}, b_{\nu j}), \tag{S138}$$
$$\ell_\tau = m_{\log x}(a_\tau, b_\tau), \qquad r_\tau = m_{x^{-1}}(a_\tau, b_\tau), \tag{S139}$$
$$\ell_\xi = m_{\log x}(a_\xi, b_\xi), \qquad r_\xi = m_{x^{-1}}(a_\xi, b_\xi). \tag{S140}$$

If ζ^2 is random, set $\ell_\zeta = m_{\log x}(a_\zeta^\star, b_\zeta^\star)$ and $r_\zeta = m_{x^{-1}}(a_\zeta^\star, b_\zeta^\star)$; if it is fixed at ζ_0^2, set $\ell_\zeta = \log \zeta_0^2$ and $r_\zeta = 1/\zeta_0^2$.

The coefficient-prior contribution is

$$\mathcal{L}_{\beta,\mathrm{hs}} = \sum_{j\in\mathcal{A}} \left\{ -\frac{1}{2}\log(2\pi) - \frac{1}{2}\ell_\tau - \frac{1}{2}\ell_{\lambda j} - \frac{1}{2}\bar{\beta}_j^2 r_\tau r_{\lambda j} \right\}, \tag{S141}$$
$$\mathcal{L}_{\beta,\mathrm{slab}} = \sum_{j\in\mathcal{A}} \left\{ -\frac{1}{2}\log(2\pi) - \frac{1}{2}\ell_\zeta - \frac{1}{2}\bar{\beta}_j^2 r_\zeta \right\}. \tag{S142}$$

The local, global, and slab prior contributions are

$$\mathcal{L}_{\lambda|\nu} = \sum_{j\in\mathcal{A}} \left\{ -\frac{1}{2}\ell_{\nu j} - \log\Gamma(1/2) - \frac{3}{2}\ell_{\lambda j} - r_{\nu j} r_{\lambda j} \right\}, \tag{S143}$$

$$\mathcal{L}_{\nu} = \sum_{j\in\mathcal{A}} \left\{ -\log\Gamma(1/2) - \frac{3}{2}\ell_{\nu j} - r_{\nu j} \right\}, \tag{S144}$$

$$\mathcal{L}_{\tau|\xi} = -\frac{1}{2}\ell_{\xi} - \log\Gamma(1/2) - \frac{3}{2}\ell_{\tau} - r_{\xi} r_{\tau}, \tag{S145}$$

$$\mathcal{L}_{\xi} = \frac{1}{2}\log(\tau_0^{-2}) - \log\Gamma(1/2) - \frac{3}{2}\ell_{\xi} - \tau_0^{-2} r_{\xi}, \tag{S146}$$

$$\mathcal{L}_{\zeta} = \begin{cases} a_\zeta \log b_\zeta - \log\Gamma(a_\zeta) - (a_\zeta + 1)\ell_\zeta - b_\zeta r_\zeta, & \zeta^2 \text{ random}, \\ 0, & \zeta^2 \text{ fixed}. \end{cases} \tag{S147}$$

The entropy contribution is

$$\mathcal{L}_{\text{ent}} = \sum_{j\in\mathcal{A}} h_{\text{IG}}(a_{\lambda j}, b_{\lambda j}) + \sum_{j\in\mathcal{A}} h_{\text{IG}}(a_{\nu j}, b_{\nu j}) + h_{\text{IG}}(a_\tau, b_\tau) + h_{\text{IG}}(a_\xi, b_\xi) + \mathbb{I}(\zeta^2 \text{ random}) h_{\text{IG}}(a_\zeta^\star, b_\zeta^\star). \tag{S148}$$

Finally, when the intercept is not included in $\mathcal{A}$, the package adds the fixed Gaussian intercept-prior contribution

$$\mathcal{L}_{\beta_0} = \frac{1}{2}\{\log(\pi_0) - \log(2\pi)\} - \frac{1}{2}\pi_0 \bar{\beta}_0^2, \tag{S149}$$

where π_0 is the fixed intercept precision. If the intercept is shrunk, this term is omitted because the intercept is already represented in the active `rhs_ns` block.

Thus the package contribution is

$$\mathcal{L}_{\text{rhs\_ns}} = \mathcal{L}_{\beta,\text{hs}} + \mathcal{L}_{\beta,\text{slab}} + \mathcal{L}_{\lambda|\nu} + \mathcal{L}_{\nu} + \mathcal{L}_{\tau|\xi} + \mathcal{L}_{\xi} + \mathcal{L}_{\zeta} + \mathcal{L}_{\text{ent}} + \mathcal{L}_{\beta_0}. \tag{S150}$$

All terms are available in closed form because each scale factor is inverse-gamma. This is the `rhs_ns` component of the full static-model ELBO; the likelihood, latent-variable, and (σ, γ) terms are added by the surrounding static AL or exAL variational routine.

S8. Forecasting, diagnostics, and posterior-predictive synthesis

S8.1. Forecasting

For a posterior draw m at forecast origin $\tilde{t}$, the state forecast recursion in the main article gives

$$\boldsymbol{\theta}_{\tilde{t}+k}^{(m)} \sim \mathcal{N}\{a_{\tilde{t}}^{(m)}(k), R_{\tilde{t}}^{(m)}(k)\}, \qquad k = 1, \ldots, K. \tag{S151}$$

Given a forecasted state draw, posterior predictive draws are generated from

$$Y_{\tilde{t}+k}^{\text{rep},(m)} \sim \text{exAL}_{p_0}\{\mathbf{F}_{\tilde{t}+k}^\top \boldsymbol{\theta}_{\tilde{t}+k}^{(m)}, \sigma^{(m)}, \gamma^{(m)}\}. \tag{S152}$$

For the AL/DQLM special case, set $\gamma^{(m)} = 0$. The function `exdqlmForecast()` implements this recursion for MCMC and LDVB fitted objects; for LDVB fits, the posterior draws are replaced by draws from the fitted variational factors.

S8.2. Diagnostics

The dynamic diagnostics use the plug-in MAP standardized one-step-ahead forecast-error sequence stored as `map.standard.forecast.errors`. If $\widehat{f}_t$ and $\widehat{q}_t$ denote the fitted one-step-ahead location and variance used in this sequence, the diagnostic error and its normal-score transform are

$$\widehat{e}_t = \frac{y_t - \widehat{f}_t}{\sqrt{\widehat{q}_t}}, \qquad \widehat{u}_t = \Phi(\widehat{e}_t). \tag{S153}$$

The QQ plot, ACF plot, and standardized forecast-error plot are computed from $\{\widehat{e}_t\}$ or $\{\widehat{u}_t\}$ rather than from posterior uncertainty bands for residuals. The reported KL summaries are nearest-neighbor estimates of the divergence between this empirical standardized-error sample and a reference standard-normal sample supplied through `ref` or generated internally; the flipped version reverses the two arguments.

For predictive draws $y_t^{(1)}, \ldots, y_t^{(M)}$, the empirical CRPS estimator is

$$\widehat{\mathrm{CRPS}}_t = \frac{1}{M}\sum_{m=1}^{M} |y_t^{(m)} - y_t^{\mathrm{obs}}| - \frac{1}{2M^2}\sum_{m=1}^{M}\sum_{\ell=1}^{M} |y_t^{(m)} - y_t^{(\ell)}|. \tag{S154}$$

The reported CRPS is the average of (S154) over the diagnostic time points (Gneiting and Raftery 2007). For target quantile p_0, the posterior predictive loss criterion returned as `pplc` is

$$\widehat{\mathrm{PPLC}} = \sum_t \frac{1}{M}\sum_{m=1}^{M} \rho_{p_0}\{y_t^{\mathrm{obs}} - y_t^{\mathrm{rep},(m)}\}, \tag{S155}$$

where ρ_{p_0} is the check-loss function (Gelfand and Ghosh 1998). The dynamic diagnostic routine returns these summaries. The static diagnostic routine returns check-loss summaries and reference-quantile errors when a reference quantile is supplied.

S8.3. Posterior-predictive synthesis

The function `quantileSynthesis()` combines posterior predictive draws from separately fitted models at ordered levels $p_1 < \cdots < p_K$. For each time point and draw index, it forms the quantile-specific predictive curve, applies the requested monotonicity correction when fitted quantiles cross, and interpolates across p to obtain draws from one posterior predictive distribution. The monotonicity correction uses isotonic adjustment and, when requested, global monotone rearrangement (Barlow, Bartholomew, Bremner, and Brunk 1972; Chernozhukov, Fernández-Val, and Galichon 2010).

Algorithm S8. Posterior-predictive synthesis.

Input: posterior predictive draws or fitted forecast objects from models at ordered quantile levels $p_1 < \cdots < p_K$, together with the requested monotonicity correction and interpolation settings.

Output: synthesized posterior predictive draws and summary intervals on the common predictive scale.

(1) Extract or collect posterior predictive draw matrices for each fitted quantile level.

(2) For each time point and retained draw index, form the quantile-specific predictive curve over $p_1, \ldots, p_K$.

(3) Apply the requested monotonicity correction.

(4) Interpolate across quantile levels.

(5) Return synthesized posterior predictive draws and summary intervals.

S9. Numerical and ELBO implementation blocks

S9.1. Reusable moment and entropy formulas

The variational routines monitor

$$\mathcal{L}(q) = \mathbb{E}_q\{\log p(\text{complete data})\} + H(q), \qquad H(q) = -\mathbb{E}_q\{\log q\}. \tag{S156}$$

The following moment and entropy formulas are reused by the dynamic and static algorithms. If $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{m}, \mathbf{V})$ with $\boldsymbol{x} \in \mathbb{R}^d$, then

$$H\{\mathcal{N}(\boldsymbol{m}, \mathbf{V})\} = \frac{1}{2}\{d(1 + \log 2\pi) + \log|\mathbf{V}|\}. \tag{S157}$$

If $x \sim \mathrm{IG}(a, b)$ under the parameterization in Section S2, then

$$\mathbb{E}(x^{-1}) = \frac{a}{b}, \qquad \mathbb{E}(\log x) = \log b - \psi(a), \tag{S158}$$

and

$$H\{\mathrm{IG}(a, b)\} = a + \log b + \log\Gamma(a) - (a + 1)\psi(a), \tag{S159}$$

where $\psi(\cdot)$ is the digamma function.

If $x \sim \mathrm{GIG}(\lambda, \chi, \psi_{\mathrm{GIG}})$ and $z = (\chi\psi_{\mathrm{GIG}})^{1/2}$, let $m_x = \mathbb{E}(x)$, $m_{1/x} = \mathbb{E}(x^{-1})$, and $m_{\log x} = \mathbb{E}(\log x)$. Then

$$\begin{aligned} H\{\mathrm{GIG}(\lambda, \chi, \psi_{\mathrm{GIG}})\} = \log\{2K_\lambda(z)\} - \frac{\lambda}{2}\log\left(\frac{\psi_{\mathrm{GIG}}}{\chi}\right) - (\lambda - 1)m_{\log x} \\ + \frac{1}{2}\{\chi m_{1/x} + \psi_{\mathrm{GIG}} m_x\}. \end{aligned} \tag{S160}$$

The moments m_x and $m_{1/x}$ follow from (S11); $m_{\log x}$ is the derivative of $\log K_\lambda(z)$ with respect to λ plus $\frac{1}{2}\log(\chi/\psi_{\mathrm{GIG}})$, and is evaluated numerically in the package.

If $x \sim \mathcal{N}^+(m, V)$, let $a = V^{1/2}$, $z = m/a$, and $\Lambda(z) = \phi(z)/\Phi(z)$. The entropy is

$$H\{\mathcal{N}^+(m, V)\} = \frac{1}{2}\log(2\pi V) + \log\Phi(z) + \frac{1}{2}\{1 - z\Lambda(z)\}. \tag{S161}$$

This closed form is used for the truncated-normal latent factors; numerically, the package evaluates the required moments using stable tail-probability guards.

For exAL variational inference, the nonconjugate (σ, γ) block is handled on transformed coordinates

$$\ell_\sigma = \log\sigma, \qquad z_\gamma = \mathrm{logit}\left(\frac{\gamma - L}{U - L}\right). \tag{S162}$$

If $\boldsymbol{z} = (z_\gamma, \ell_\sigma)^\top$ has Laplace approximation $q_z(\boldsymbol{z}) = \mathcal{N}(\hat{\boldsymbol{z}}, \boldsymbol{\Sigma}_z)$ and $(\gamma, \sigma) = T(\boldsymbol{z})$, then the entropy contribution on the original scale is approximated as

$$H\{q(\sigma,\gamma)\} \approx \frac{1}{2}\{2(1+\log 2\pi) + \log|\boldsymbol{\Sigma}_z|\} + \mathbb{E}_{q_z}\{\log|J_T(\boldsymbol{z})|\}, \tag{S163}$$

where

$$\log|J_T(\boldsymbol{z})| = \ell_\sigma + \log(U-L) + \log r + \log(1-r), \qquad r = \operatorname{expit}(z_\gamma). \tag{S164}$$

The final expectation in (S163), and other smooth expectations of functions of (σ, γ), are evaluated by the same second-order delta approximation used in the LDVB updates.

S9.2. Bounded univariate slice sampler

Let $h(x)$ be an unnormalized log-density on a finite interval $[a, b]$, and let $x_0 \in (a, b)$. The bounded slice update implemented in the package is:

(1) Draw $e \sim \operatorname{Exp}(1)$ and set the log-slice height $z = h(x_0) - e$.

(2) Draw $u \sim \operatorname{Unif}(0, w)$ and initialize $L = x_0 - u$, $R = x_0 + w - u$.

(3) Step out left and right by increments of w, respecting the bounds a and b, until the interval endpoints lie below the slice or the maximum number of expansions is reached.

(4) Truncate the interval to $[a, b]$.

(5) Repeatedly draw $x_1 \sim \operatorname{Unif}(L, R)$. If $h(x_1) \geq z$, accept x_1. Otherwise shrink the interval to $[L, x_1]$ if $x_1 > x_0$ and to $[x_1, R]$ otherwise.

For dynamic and static exAL MCMC fits, this sampler is applied to the one-dimensional γ target on (L, U) by default.

S9.3. Laplace–delta approximation

For a nonconjugate block $\boldsymbol{\eta} \in \mathbb{R}^d$ with log target $\ell(\boldsymbol{\eta})$, let $\hat{\boldsymbol{\eta}}$ be a local mode and $\mathbf{H} = \nabla^2 \ell(\hat{\boldsymbol{\eta}})$. If $-\mathbf{H}$ is positive definite, the Laplace approximation is

$$q(\boldsymbol{\eta}) \approx \mathcal{N}\left(\hat{\boldsymbol{\eta}},\ [-\mathbf{H}]^{-1}\right). \tag{S165}$$

For a smooth function $r(\boldsymbol{\eta})$, the delta approximation used by LDVB is

$$\mathbb{E}\{r(\boldsymbol{\eta})\} \approx r(\hat{\boldsymbol{\eta}}) + \frac{1}{2}\operatorname{tr}\left[\nabla^2 r(\hat{\boldsymbol{\eta}})[-\mathbf{H}]^{-1}\right]. \tag{S166}$$

This is used for smooth functions of (σ, γ) in dynamic and static exAL models.

S10. Package-to-algorithm map

Sections S3–S9 define the targets used by the exported fitting and post-processing functions. Some routines temporarily hold or schedule selected blocks during early iterations for numerical stability. These controls affect update timing only; they do not change the posterior target, full conditional distributions, or coordinate optima listed in this supplement.

Function	Section	Derivation or computational block
`exdqlmMCMC()`	S3–S4	Dynamic DQLM/exDQLM MCMC; Algorithms S1 and S3.
`exdqlmLDVB()`	S3–S4	Dynamic DQLM/exDQLM LDVB; Algorithms S2 and S4.
`exdqlmTransferMCMC()`	S5	Transfer-function state augmentation; Algorithm S5 with Algorithm S3.
`exdqlmTransferLDVB()`	S5	Transfer-function state augmentation; Algorithm S5 with Algorithm S4.
`exalStaticMCMC()`	S6–S7	Static AL/exAL MCMC; Algorithm S6.
`exalStaticLDVB()`	S6–S7	Static AL/exAL LDVB; Algorithm S7.
`exdqlmForecast()`	S8.1	Forecast recursion and predictive draws.
`exdqlmDiagnostics()`	S8.2	MAP one-step-ahead diagnostic sequence, CRPS, and `pplc`.
`exalStaticDiagnostics()`	S8.2	Static check-loss and reference-quantile diagnostics.
`quantileSynthesis()`	S8.3	Posterior-predictive synthesis; Algorithm S8.
`dexal()`, `pexal()` `qexal()`, `rexal()`	S2	exAL distribution evaluation and simulation.
`get_gamma_bounds()`	S2	Roots L, U for a fixed p_0.

Table S4: Mapping between exported **exdqlm** functions and the derivation or computational details documented in this supplement.

References

Barata R, Prado R, Sansó B (2022). "Fast inference for time-varying quantiles via flexible dynamic models with application to the characterization of atmospheric rivers." *The Annals of Applied Statistics*, **16**(1), 247–271. doi:10.1214/21-AOAS1497.

Barlow RE, Bartholomew DJ, Bremner JM, Brunk HD (1972). *Statistical Inference Under Order Restrictions: The Theory and Application of Isotonic Regression*. John Wiley & Sons, New York.

Carter CK, Kohn R (1994). "On Gibbs sampling for state space models." *Biometrika*, **81**(3), 541–553.

Chernozhukov V, Fernández-Val I, Galichon A (2010). "Quantile and Probability Curves Without Crossing." *Econometrica*, **78**(3), 1093–1125. doi:10.3982/ECTA7880.

Frühwirth-Schnatter S (1994). "Data augmentation and dynamic linear models." *Journal of time series analysis*, **15**(2), 183–202.

Gelfand AE, Ghosh SK (1998). "Model choice: a minimum posterior predictive loss approach." *Biometrika*, **85**(1), 1–11.

Gneiting T, Raftery AE (2007). "Strictly Proper Scoring Rules, Prediction, and Estimation." *Journal of the American Statistical Association*, **102**(477), 359–378. doi:10.1198/016214506000001437.

Gonçalves K, Migon HS, Bastos LS (2017). "Dynamic quantile linear model: a Bayesian approach." *arXiv preprint arXiv:1711.00162*.

Kozumi H, Kobayashi G (2011). "Gibbs sampling methods for Bayesian quantile regression." *Journal of statistical computation and simulation*, **81**(11), 1565–1578.

Nishimura A, Suchard MA (2023). "Shrinkage with shrunken shoulders: Gibbs sampling shrinkage model posteriors with guaranteed convergence rates." *Bayesian Analysis*, **18**(2), 367–390. doi:10.1214/22-BA1308.

Wang C, Blei DM (2013). "Variational Inference in Non-Conjugate Models." *arXiv preprint arXiv:1209.4360*. URL https://arxiv.org/abs/1209.4360.

West M, Harrison PJ, Migon HS (1985). "Dynamic generalized linear models and Bayesian forecasting." *Journal of the American Statistical Association*, **80**(389), 73–83.

Yan Y, Zheng X, Kottas A (2025). "A New Family of Error Distributions for Bayesian Quantile Regression." *Bayesian Analysis*. doi:10.1214/25-BA1507.

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